



NGUYEN THANH DAT

Backend Developer (NestJS / Node.js · PHP / Laravel)

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PROFESSIONAL SUMMARY

- Nearly **5 years** of backend software engineering experience, growing from Fresher to a **Backend Developer with senior-level ownership** across ERP, MES, healthcare, e-commerce, and booking systems, mostly for Japanese clients.
- **Core expertise in NestJS/TypeScript and PHP/Laravel**, with TypeORM, MySQL/PostgreSQL query & index optimization, RESTful API design, RabbitMQ and Redis, Docker, and event-driven/queue-based processing.
- Track record of resolving **high-impact performance bottlenecks** through root-cause query/index analysis — all of them inside the NestJS ERP service, including a reporting query from 26s to 250ms and CRM query times from up to 18s down to under 200ms.
- Comfortable learning new technologies independently and diagnosing root causes before implementing fixes; involved in a project from inception (architecture & tech-stack decisions) and in optimizing team workflow for a 5-person backend team.
- Worked directly with overseas (Japanese) clients in cross-functional teams ranging from 5 to 22 members.

TECHNICAL SKILLS

Languages: TypeScript (2 yrs) · JavaScript (2 yrs) · PHP (5 yrs)

Backend Frameworks: NestJS (1 yr) · Laravel (5 yrs) · Lumen (1 yr)

Frontend: Vue.js (3 yrs) · Tailwind CSS

Architecture: Modular Architecture · RESTful API Design · Queue-Based / Async Processing (RabbitMQ) · Scheduled Batch Jobs · Cross-System Data Sync · OAuth2 / JWT (Passport) · SSO (Keycloak)

Database: MySQL (3 yrs) · PostgreSQL (3 yrs) · TypeORM · Database Design · Indexing · Query Optimization

Caching & Performance: Redis · Caching Strategies · Performance Optimization

Infrastructure: Docker · Linux · AWS S3

Testing, Monitoring & API Docs: Vitest · Unit & Feature Testing · Swagger · Winston · Laravel Telescope

Integrations: Stripe API · Firebase Cloud Messaging · Pusher · Laravel Reverb · Google Maps API · SNS Login

Tools: Git · GitHub · GitLab · Jira · Redmine · Trello

PROFESSIONAL EXPERIENCE

BEAR ACER COMPANY — Backend Developer

Aug 2025 – Present

- Sole developer on the ERP service — NestJS with TypeORM on the server, Vue on the client — aggregating data from the B2B (15-member team) and MES (7-member team) PHP systems for daily/monthly reporting, customer support, and email handling, plus hands-on contributions to MES features.
- Contributed to a **modular backend architecture** supporting multiple business domains for long-term maintainability.
- Diagnosed and resolved 3 major performance bottlenecks inside the NestJS ERP codebase through root-cause query and index analysis: replaced an on-the-fly monthly aggregation with a monthly **pre-computed report table (26s→250ms)**, cutting rows scanned from ~5M to ~380K, alongside rebuilding a composite index whose column order was wrong; added the **missing index** on the multi-column filter (**9s→80ms**); and reworked 3 CRM endpoints, splitting the worst — a LEFT JOIN across ~18 tables, including contacts at 1M and contact info at 2M rows — into a fast main-record call plus a deferred contract call (**6.1s/18.2s/8.8s → 187ms/64ms/6ms**).
- Diagnosed a **CPU/RAM contention issue** between batch jobs and the app server sharing one container on an inherited codebase with no existing monitoring — found via **`docker stats`** after noticing server slowdowns; split batch processing into a **dedicated worker container** with resource limits (**0.5 CPU / 512MB**) and self-throttling logic (Node.js CPU-usage checks) to stabilize the app server.
- Wired the cross-system sync: B2B writes each day's business data into a shared sync database, then twelve scheduled ERP batch jobs and three **RabbitMQ consumers** pull it across and recompute the ERP summaries.
- Used **Laravel Telescope** for debugging and monitoring on the B2B/MES Laravel services, alongside a separate monitoring setup for the NestJS service.
- Designed and built the **MES Billing module**: multi-case billing calculation logic, PDF invoice generation (re-mapped on edit/re-download), multi-recipient email sending, billing list, and role-based permissions.

- Implemented real-time inventory updates using **Redis, S3, and Laravel Reverb**; wrote unit and feature tests for new features.

Tech Stack: NestJS, TypeORM, Vue.js, MySQL, PostgreSQL, RabbitMQ, Redis, Docker, Vitest, Laravel, S3, Linux

SOTATEK COMPANY — Backend Developer

Mar 2025 – Aug 2025 | 16-member team, 5-person backend team

- Joined a **hospital management system project from inception**, deciding technical solutions and the initial architecture and technology stack for the 5-person backend team.
- Redesigned the backend team's **Git workflow and release process**: replaced ad-hoc "dev-environment" releases with feature branches checked out from main and **sprint-scoped planning**, so QC could plan test cases per sprint and only QC-verified features shipped — sharply reducing production bugs and regressions.
- Introduced a **peer-review checklist** for the team and later an **AI-assisted PR review** step, reducing review overhead for out-of-scope changes.
- Independently identified and fixed **missing indexes and inefficient queries** to improve backend performance.
- Implemented secure authentication via **OAuth (Laravel Passport)** and introduced **Redis caching**.

Tech Stack: Lumen, Laravel, MySQL, Redis, Docker, S3, OAuth2/Laravel Passport, Firebase Cloud Messaging, Swagger, Linux

KOZOCOM COMPANY — Backend Developer (Middle-level)

Feb 2024 – Mar 2025 | 16-member team

- Worked independently within a **Scrum team**, following clean-code and clean-architecture practices — a period of significant technical growth.
- Developed and maintained backend features for a **warehouse/inventory management platform** (Japanese client) using Laravel and PostgreSQL.
- Built a **legacy-to-new-system data migration** via CSV import handling approximately **2–3 million records per import**.
- Designed database schemas and implemented business logic for purchasing/selling and store-management modules, with a Redis queue handling data sync across the platform.
- Added **feature and unit tests** to the backend workflow.
- Reviewed code and supported junior developers in completing technical tasks.

Tech Stack: Laravel, PostgreSQL, Swagger, Redis, Pusher, Linux

FLYDINO TECHNOLOGY COMPANY — Junior Backend Developer

Apr 2023 – Feb 2024

- Independently developed backend features (Laravel) and frontend functionality (Vue 3 + TypeScript / Tailwind) across multiple client projects, including a project-management platform and a booking app, while strengthening technology-selection judgment.
- Designed database structures for new features and integrated Keycloak SSO / SNS login and S3/FTP storage on a recruitment platform (18-member team).
- Reviewed and supported junior developers' code.

Tech Stack: Laravel, Vue 3 + TypeScript, Tailwind, MySQL, S3, Keycloak, Google Maps API

NINE PLUS SOFTWARE DEVELOPMENT COMPANY — Fresher Backend Developer

Oct 2021 – Apr 2023

- Started as a Fresher, learning Laravel/PHP fundamentals and RESTful API development through hands-on feature implementation for an HR management system (Japanese client, 10-member team).
- Designed database schemas and implemented business logic for recruitment and HR modules, and later for an e-commerce platform with Stripe payment integration.
- Maintained and refactored the existing codebase.

Tech Stack: Laravel, MySQL, S3, Stripe API, Docker

SELECTED PROJECTS

B2B ERP & MES Platform (Bear Acer)

Aug 2025 – Present | Japanese Client | Sole developer on the ERP service

- Sole developer on the ERP service (NestJS + TypeORM, Vue on the client) inside an enterprise ERP + MES platform covering Accounting, CRM, HR, Billing, Order, Shipping, and Inventory management.
- Diagnosed and resolved 3 major query/index bottlenecks (**26s→250ms, 9s→80ms, up to 18.2s→64ms**); split a batch process into a resource-limited worker container (**0.5 CPU / 512MB**) with self-throttling to fix CPU/RAM contention with the app server.

Tech Stack: NestJS, TypeORM, Vue.js, MySQL, PostgreSQL, RabbitMQ, Redis, Docker

Hospital Management System (Sotatek)

Mar 2025 – Aug 2025 | Japanese Client | 16-member team, 5-person backend team

- Comprehensive hospital management platform: appointment booking, course/shift/contract management, and secure **OAuth (Laravel Passport)** authentication.
- On the backend team from project inception; redesigned the team's Git workflow and release process (sprint-scoped feature branches, QC-gated releases), sharply reducing production bugs.

Tech Stack: Lumen, Laravel, MySQL, Redis, Docker, Firebase Cloud Messaging

AI Skin Analysis & Booking Platform ("SBea")

Dec 2023 – Feb 2024 | Japanese Client | 8-member team

- Integrated the **ChatGPT-4 API** to analyze uploaded facial photos and generate skin-condition assessments, feeding results into a personalized product-recommendation flow.
- Built the service-booking, store-management, and authentication/authorization features end-to-end.

Tech Stack: Laravel (MVC), JavaScript, Tailwind, MySQL

Project Management & Recruitment Platform ("En")

Jun 2023 – Dec 2023 | Japanese Client | 18-member team

- Implemented backend APIs and **role-based authorization workflows** for project, contract, and task-management modules.
- Integrated **Keycloak SSO**, SNS authentication, and S3/FTP storage for enterprise-grade access control and file handling.

Tech Stack: Laravel (RESTful API), Vue 3 + TypeScript, Tailwind, MySQL, Swagger, S3, Keycloak

Driver Booking Platform ("Viet")

Apr 2023 – Jun 2023 | Vietnamese Client | 8-member team

- Implemented **real-time distance and fare calculation** between pickup and destination points using the Google Maps API.
- Built driver booking, rating, and driver/user management features.

Tech Stack: Laravel (RESTful API), Vue 3 + TypeScript, Tailwind, Google Maps API, MySQL

E-commerce & Payment Integrations — 3 platforms

2022 – 2025 | Japanese & Vietnamese clients | 6–20 member teams

- Built multi-seller store, product and order-management features with payment and shipping workflows for a Japanese client (Count, 2022, 12–20 members).
- Integrated the **Stripe** payment gateway on all three, alongside SNS login and Redis catalog caching. The other two: a covid test-kit sales platform with order management and CSV import/export (Sell Kits Test, 2022–2023, 9 members), and a company-management platform with job management and business-plan trading (AOI, 2024–2025, 6 members).

Tech Stack: Laravel (RESTful API & MVC), MySQL, Stripe API, Redis, Docker, S3, JavaScript, Tailwind

PERSONAL PROJECT

MIA-POS — Self-funded Side Project (in testing)

Ongoing | Preview build: mia-store-remote.pages.dev

- Self-directed and self-funded POS/online-store platform, built toward a future one-person business.
- Structured into domain-oriented services (**auth, admin, store, payment**) behind a Caddy gateway, split by language: Laravel for auth, admin and payments; Go with Gin for stores, inventory and the log worker; Python with FastAPI for the AI features; three Vue 3 apps in TypeScript on a Bun workspace. Services still share one database — a modular monolith, a deliberate trade-off at this stage.
- Set up **CI/CD via GitHub Actions** for automated testing and deployment.

EDUCATION

FPT Polytechnic — Software Application | Jul 2019 – Dec 2021